A truncated elementary Selberg sieve of Pintz

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This post is a continuation of the previous post on sieve theory, which is an ongoing part of the Polymath8 project. As the previous post was getting somewhat full, we are rolling the thread over to the current post.

In this post we will record a new truncation of the elementary Selberg sieve discussed in this previous post (and also analysed in the context of bounded prime gaps by Graham-Goldston-Pintz-Yildirim and Motohashi-Pintz) that was recently worked out by Janos Pintz, who has kindly given permission to share this new idea with the Polymath8 project. This new sieve decouples the δ parameter that was present in our previous analysis of Zhang's argument into two parameters, a quantity δ that used to measure smoothness in the modulus, but now measures a weaker notion of "dense divisibility" which is what is really needed in the Elliott-Halberstam type estimates, and a second quantity δ' which still measures smoothness but is allowed to be substantially larger than δ. Through this decoupling, it appears that the κ type losses in the sieve theoretic part of the argument can be almost completely eliminated (they basically decay exponential in δ' and have only mild dependence on δ, whereas the Elliott-Halberstam analysis is sensitive only to δ, allowing one to set δ far smaller than previously by keeping δ' large). This should lead to noticeable gains in the k_0 quantity in our analysis.

To describe this new truncation we need to review some notation. As in all previous posts (in particular, the first post in this series), we have an asymptotic parameter x going off to infinity, and all quantities here are implicitly understood to be allowed to depend on x (or to range in a set that depends on x) unless they are explicitly declared to be *fixed*. We use the usual asymptotic notation $O(), o(), \ll$ relative to this parameter x. To be able to ignore local factors (such as the singular series $\mathfrak{G}$), we also use the "W-trick" (as discussed in the first post in this series): we introduce a parameter w that grows very slowly with x, and set $W := \prod_{p<w} p$.

For any fixed natural number k_0, define an *admissible k_0-tuple* to be a fixed tuple $\mathcal{H}$ of k_0 distinct integers which avoids at least one residue class modulo p for each prime p. Our objective is to obtain the following conjecture $DHL[k_0, 2]$ for as small a value of the parameter k_0 as possible:

> **Conjecture 1** ($DHL[k_0, 2]$) Let $\mathcal{H}$ be a fixed admissible k_0-tuple. Then there exist infinitely many translates $n + \mathcal{H}$ of $\mathcal{H}$ that contain at least two primes.

The twin prime conjecture asserts that $DHL[k_0, 2]$ holds for k_0 as small as 2, but currently we are only able to establish this result for $k_0 \geq 6329$ (see this comment). However, with the new truncated sieve of Pintz described in this post, we expect to be able to lower this threshold $k_0 \geq 6329$ somewhat.

In previous posts, we deduced $DHL[k_0,2]$ from a technical variant $MPZ[\varpi,\delta]$ of the Elliot-Halberstam conjecture for certain choices of parameters $0 < \varpi < 1/4$, $0 < \delta < 1/4 + \varpi$. We will use the following formulation of $MPZ[\varpi,\delta]$:

Conjecture 2 ($MPZ[\varpi,\delta]$) Let $\mathcal{H}$ be a fixed k_0-tuple (not necessarily admissible) for some fixed $k_0 \geq 2$, and let $b\ (W)$ be a primitive residue class. Then

$$\sum_{q \in \mathcal{S}_I: q < x^{1/2+2\varpi}} \sum_{a \in C(q)} \left|\Delta_{b,W}(\Lambda; q, a)\right| = O(x \log^{-A} x) \tag{1}$$

for any fixed $A > 0$, where $I = (w, x^\delta)$, $\mathcal{S}_I$ are the square-free integers whose prime factors lie in I, and $\Delta_{b,W}(\Lambda; q, a)$ is the quantity

$$\Delta_{b,W}(\Lambda; q, a) := \Big| \sum_{x \leq n \leq 2x: n = b\ (W); n = a\ (q)} \Lambda(n) - \frac{1}{\phi(q)} \sum_{x \leq n \leq 2x: n = b\ (W)} \Lambda(n) \Big| . \tag{2}$$

and $C(q)$ is the set of congruence classes

$$C(q) := \{a \in (\mathbf{Z}/q\mathbf{Z})^\times : P(a) = 0\}$$

and P is the polynomial

$$P(a) := \prod_{h \in \mathcal{H}} (a + h).$$

The conjecture $MPZ[\varpi,\delta]$ is currently known to hold whenever $87\varpi + 17\delta < \frac{1}{4}$ (see this comment and this confirmation). Actually, we can prove a stronger result than $MPZ[\varpi,\delta]$ in this regime in a couple ways. Firstly, the congruence classes $C(q)$ can be replaced by a more general system of congruence classes obeying a certain controlled multiplicity axiom; see this post. Secondly, and more importantly for this post, the requirement that the modulus q lies in $\mathcal{S}_I$ can be relaxed; see below.

To connect the two conjectures, the previously best known implication was the folowing (see Theorem 2 from this post):

Theorem 3 Let $0 < \varpi < 1/4$, $0 < \delta < 1/4 + \varpi$ and $k_0 \geq 2$ be such that we have the inequality

$$(1 + 4\varpi)(1 - \kappa') > \frac{j_{k_0-2}^2}{k_0(k_0-1)}(1 + \kappa) \tag{3}$$

where $j_{k_0-2} = j_{k_0-2,1}$ is the first positive zero of the Bessel function J_{k_0-2}, and $\kappa, \kappa' > 0$ are the quantities

$$\kappa := \sum_{1 \leq n < \frac{1+4\varpi}{4\delta}} \frac{3^n + 1}{2} \frac{k_0^n}{n!} \Big(\int_{4\delta/(1+4\varpi)}^1 (1-t)^{k_0/2} \frac{dt}{t} \Big)^n$$

and

$$\kappa' := \sum_{2 \leq n < \frac{1+4\varpi}{4\delta}} \frac{3^n - 1}{2} \frac{(k_0-1)^n}{n!}$$

$$(\int_{4\delta/(1+4\varpi)}^1 (1-t)^{(k_0-1)/2} \frac{dt}{t})^n.$$

Then $MPZ[\varpi, \delta]$ implies $DHL[k_0, 2]$.

Actually there have been some slight improvements to the quantities κ, κ'; see the comments to this previous post. However, the main error κ remains roughly of the order $\delta^{-1} \exp(-2k_0\delta)$, which limits one from taking δ too small.

To improve beyond this, the first basic observation is that the smoothness condition $q \in \mathcal{S}_I$, which implies that all prime divisors of q are less than x^δ, can be relaxed in the proof of $MPZ[\varpi, \delta]$. Indeed, if one inspects the proof of this proposition (described in these three previous posts), one sees that the key property of q needed is not so much the smoothness, but a weaker condition which we will call (for lack of a better term) *dense divisibility*:

Definition 4 Let $y > 1$. A positive integer q is said to be *y-densely divisible* if for every $1 \leq R \leq q$, one can find a factor of q in the interval $[y^{-1}R, R]$. We let $\mathcal{D}_y$ denote the set of positive integers that are y-densely divisible.

Certainly every integer which is y-smooth (i.e. has all prime factors at most y is also y-densely divisible, as can be seen from the greedy algorithm; but the property of being y-densely divisible is strictly weaker than y-smoothness, which is a fact we shall exploit shortly.

We now define $MPZ'[\varpi, \delta]$ to be the same statement as $MPZ[\varpi, \delta]$, but with the condition $q \in \mathcal{S}_I$ replaced by the weaker condition $q \in \mathcal{S}_{[w,+\infty)} \cap \mathcal{D}_{x^\delta}$. The arguments in previous posts then also establish $MPZ'[\varpi, \delta]$ whenever $87\varpi + 17\delta < \frac{1}{4}$.

The main result of this post is then the following implication, essentially due to Pintz:

Theorem 5 Let $0 < \varpi < 1/4$, $0 < \delta \leq \delta' < 1/4 + \varpi$, $A \geq 0$, and $k_0 \geq 2$ be such that

$$(1+4\varpi)(1 - 2\kappa_1 - 2\kappa_2 - 2\kappa_3) > \frac{j_{k_0-2}^2}{k_0(k_0-1)}$$

where

$$\kappa_1 := \int_\theta^1 (1-t)^{(k_0-1)/2} \frac{dt}{t}$$

$$\kappa_2 := (k_0-1) \int_\theta^1 (1-t)^{k_0-1} \frac{dt}{t}$$

$$\kappa_3 := e^A \frac{G_{k_0-1,\tilde{\theta}}(0,0)}{G_{k_0-1}(0,0)} \sum_{0 \leq J \leq 1/\tilde{\delta}} \frac{(k_0-1)^J}{J!} (\int_{\tilde{\delta}}^\theta e^{-At} \frac{dt}{t})^J$$

and

$$\theta := \frac{\delta'}{1/4+\varpi}$$

$$\tilde{\theta} := \frac{(\delta'-\delta)/2+\varpi}{1/4+\varpi}$$

$$\tilde{\delta} := \frac{\delta}{1/4+\varpi}$$

$$G_{k_0-1}(0,0) := \int_0^1 f(t)^2 \frac{t^{k_0-2}}{(k_0-2)!}\,dt$$

$$G_{k_0-1,\tilde{\theta}}(0,0) := \int_0^{\tilde{\theta}} f(t)^2 \frac{t^{k_0-2}}{(k_0-2)!}\,dt$$

and

$$f(t) := t^{1-k_0/2} J_{k_0-2}(\sqrt{t} j_{k_0-2}).$$

Then $MPZ'[\varpi,\delta]$ implies $DHL[k_0,2]$.

This theorem has rather messy constants, but we can isolate some special cases which are a bit easier to compute with. Setting $\delta' = \delta$, we see that κ_3 vanishes (and the argument below will show that we only need $MPZ[\varpi,\delta]$ rather than $MPZ'[\varpi,\delta]$), and we obtain the following slight improvement of Theorem 3:

Theorem 6 Let $0 < \varpi < 1/4$, $0 < \delta < 1/4+\varpi$ and $k_0 \geq 2$ be such that we have the inequality

$$(1+4\varpi)(1-2\kappa_1-2\kappa_2) > \frac{j_{k_0-2}^2}{k_0(k_0-1)} \tag{4}$$

where

$$\kappa_1 := \int_{4\delta/(1+4\varpi)}^1 (1-t)^{(k_0-1)/2}\,\frac{dt}{t}$$

$$\kappa_2 := (k_0-1)\int_{4\delta/(1+4\varpi)}^1 (1-t)^{k_0-1}\,\frac{dt}{t}.$$

Then $MPZ[\varpi,\delta]$ implies $DHL[k_0,2]$.

This is a little better than Theorem 3, because the error $2\kappa_1+2\kappa_2$ has size about $\frac{1}{2k_0\delta}\exp(-2k_0\delta) + \frac{1}{2\delta}\exp(-4k_0\delta)$, which compares favorably with the error in Theorem 3 which is about $\frac{1}{\delta}\exp(-2k_0\delta)$. This should already give a "cheap" improvement to our current threshold $k_0 \geq 6329$, though it will fall short of what one would get if one fully optimised over all parameters in the above theorem.

Returning to the full strength of Theorem 5, let us obtain a crude upper bound for κ_3 that is a little simpler to understand. Extending the J summation to infinity and using the Taylor series for the exponential, we have

$$\kappa_3 \leq \frac{G_{k_0-1,\tilde\theta}(0,0)}{G_{k_0-1}(0,0)} \exp(A + (k_0-1)\int_{\tilde\delta}^{\theta} e^{-At}\frac{dt}{t}).$$

We can crudely bound

$$\int_{\tilde\delta}^{\theta} e^{-At}\frac{dt}{t} \leq \frac{1}{A\tilde\delta}$$

and then optimise in A to obtain

$$\kappa_3 \leq \frac{G_{k_0-1,\tilde\theta}(0,0)}{G_{k_0-1}(0,0)} \exp(2(k_0-1)^{1/2}\tilde\delta^{-1/2}).$$

Because of the t^{k_0-2} factor in the integrand for G_{k_0-1} and $G_{k_0-1,\tilde\theta}$, we expect the ratio $\frac{G_{k_0-1,\tilde\theta}(0,0)}{G_{k_0-1}(0,0)}$ to be of the order of $\tilde\theta^{k_0-1}$, although one will need some theoretical or numerical estimates on Bessel functions to make this heuristic more precise. Setting $\tilde\theta$ to be something like $1/2$, we get a good bound here as long as $\tilde\delta \gg 1/k_0$, which at current values of δ, k_0 is a mild condition.

Pintz's argument uses the elementary Selberg sieve, discussed in this previous post, but with a more efficient estimation of the quantity β, in particular avoiding the truncation to moduli d between $x^{-\delta}R$ and R which was the main source of inefficiency in that previous post. The basic idea is to "linearise" the effect of the truncation of the sieve, so that this contribution can be dealt with by the union bound (basically, bounding the contribution of each large prime one at a time). This mostly avoids the more complicated combinatorial analysis that arose in the analytic Selberg sieve, as seen in this previous post.

— 1. Review of previous material —

In this section we collect some results from previous posts which we will need.

We first record an asymptotic for multiplicative functions. For any natural number k, define a *k-dimensional multiplicative function* to be a multiplicative function $f : \mathbf{N} \to \mathbf{R}$ which obeys the asymptotic

$$f(p) = k + O(\frac{1}{p})$$

for all $p > w$. The following result is Lemma 8 from this previous post:

> **Lemma 7** Let $I = (w, +\infty)$. Let k be a fixed positive integer, and let $f : \mathbf{N} \to \mathbf{R}$ be a multiplicative function of dimension k. Then for any fixed compactly supported, Riemann-integrable function $g : \mathbf{R} \to \mathbf{R}$, and any $R > x^c$ for some fixed $c > 0$, one has
>
> $$\sum_{d \in \mathcal{S}_I} \frac{f(d)}{d} g(\frac{\log d}{\log R}) = (\frac{\phi(W)}{W}\log R)^k (\int_0^\infty g(t)\frac{t^{k-1}}{(k-1)!}\,dt + o(1)).$$

Next, we record a criterion for $DHL[k_0, 2]$, which is Lemma 7 from this previous post:

> **Lemma 8 (Criterion for DHL)** Let $k_0 \geq 2$. Suppose that for each fixed admissible k_0 -tuple $\mathcal{H}$ and each congruence class $b\ (W)$ such that $b + h$ is coprime to W for all $h \in \mathcal{H}$, one can find a non-negative weight function $\nu : \mathbf{N} \to \mathbf{R}^+$, fixed quantities $\alpha, \beta > 0$, a quantity $B > 0$, and a fixed positive power R of x such that one has the upper bound
>
> $$\sum_{x \leq n \leq 2x : n = b\ (W)} \nu(n) \leq (\alpha + o(1)) B \frac{x}{W}, \tag{5}$$
>
> the lower bound
>
> $$\sum_{x \leq n \leq 2x : n = b\ (W)} \nu(n) \theta(n + h_i) \geq (\beta - o(1)) B \frac{x}{W} \log R \tag{6}$$
>
> for all $h_i \in \mathcal{H}$, and the key inequality
>
> $$\frac{\log R}{\log x} > \frac{1}{k_0} \frac{\alpha}{\beta} \tag{7}$$
>
> holds. Then $DHL[k_0, 2]$ holds. Here $\theta(n)$ is defined to equal $\log n$ when n is prime and 0 otherwise.

— 2. Pintz's argument —

We can now prove Theorem 5. Fix $\varpi, \delta, \delta', k_0$ to obey the hypotheses of this theorem. Let $b\ (W)$ be a congruence class with $b + h$ coprime to W for all $h \in \mathcal{H}$ (this class exists by the admissibility of $\mathcal{H}$). We apply Lemma 8 with

$$B := (\frac{\phi(W)}{W} \log R)^{k_0}$$

the elementary Selberg sieve $\nu = \nu_{\mathcal{X}}$ defined by

$$\nu(n) := (\sum_{d \in \mathcal{S}_{(w, +\infty)} : d \mid P(n)} \mu(d) a_d)^2$$

where

$$a_d := \frac{1}{\Phi(d)\Delta(d)} \sum_{q \in \mathcal{S}_{(w, +\infty)} : (q,d) = 1; qd \in \mathcal{X}} \frac{1}{\Phi(q)} f'(\frac{\log dq}{\log R}), \tag{8}$$

Φ, Δ are the multiplicative functions

$$\Phi(d) := \prod_{p \mid d} \frac{p - k_0}{k_0}$$

and

$$\Delta(d) := \prod_{p \mid d} \frac{k_0}{p},$$

the sieve level R is given by the formula

$$R := x^{1/4+\varpi},$$

$f : \mathbf{R} \to \mathbf{R}$ is a fixed smooth function supported on $[-1,1]$, and $\mathcal{X}$ is a certain subset of $\mathcal{S}_{(w,+\infty)}$ to be chosen shortly. (The constraints $q \in \mathcal{S}_{(w,+\infty)}$ and $(q,d) = 1$ are redundant given that $qd \in \mathcal{X}$, but we retain them for emphasis.) We will assume that f non-negative and non-increasing on $[0,1]$. In this previous post, we considered this sieve with $\mathcal{X}$ equal to either all of $\mathcal{S}_{(w,+\infty)}$, or the subset $\mathcal{S}_{(w,x^\delta)}$ consisting of smooth numbers. For now, we will discuss the estimates as far as we can without having to explicitly specify $\mathcal{X}$.

We first consider the asymptotic (5). By arguing exactly as in Section 2 (or Section 3) of this previous post, we can write the left-hand side of (5), up to errors of $o(B\frac{x}{W})$, as

$$\sum_{d_0 \in \mathcal{X}} \frac{1}{\Phi(d_0)} f'(\frac{\log d_0}{\log R})^2.$$

The summand here is non-negative, so we may crudely replace $\mathcal{X}$ by all of $\mathcal{S}_{(w,+\infty)}$ and apply Lemma 7 to obtain (5) with

$$\alpha := \int_0^1 f'(t)^2 \frac{t^{k_0-1}}{(k_0-1)!}\, dt.$$

Now we turn to the more difficult asymptotic (6). The left-hand side expands as

$$\sum_{d_1,d_2 \in \mathcal{S}_I} \mu(d_1) a_{d_1} \mu(d_2) a_{d_2} \sum_{x \leq n \leq 2x : [d_1,d_2] \mid P(n); n = b\ (W)} \theta(n+h_i).$$

As observed in Section 2 of this previous post, we have

$$\sum_{x \leq n \leq 2x : [d_1,d_2] \mid P(n); n = b\ (W)} \theta(n+h) = \frac{1}{\phi(W)} x \Delta^*([d_1,d_2]) + O(E^*([d_1,d_2]))$$

where

$$\Delta^*(q) := \prod_{p \mid q} \frac{k_0-1}{p-1}$$

and

$$E^*(q) = \sum_{a \in C_i(q)} \Big| \sum_{x \leq n \leq 2x : n = b\ (W); n = a\ (q)} \theta(n) - \frac{x}{\phi(Wq)} \Big|.$$

Now let $\epsilon > 0$ be a small fixed constant to be chosen later, and suppose the following claim holds:

> **Claim 1 (Dense divisibility of moduli)** Whenever $q = [d_1,d_2]$ and a_{d_1}, a_{d_2} are non-zero, then either $q \leq x^{1/2-\epsilon}$ or else $q \in \mathcal{D}_{x^\delta}$.

Then from the Bombieri-Vinogradov theorem (for the $q \leq x^{1/2-\epsilon}$ moduli) or the hypothesis $MPZ'[\varpi,\delta]$ (for the larger moduli, noting that $q \leq R^2 = x^{1/2+2\varpi}$) and standard arguments (cf. Proposition 5 of this post) we have

$$\sum_q h(q) E^*(q) \ll x \log^{-A} x$$

for any fixed $A > 0$ and any multiplicative function h of a fixed dimension k, where q ranges only over those integers of the form $q = [d_1, d_2]$ with a_{d_1}, a_{d_2} non-zero. From this we easily see (arguing as in Section 2 of this previous post) that the contribution of the E^* error term is $o(B \frac{x}{W} \log R)$, and we are left with establishing the lower bound

$$\sum_{d_1, d_2 \in \mathcal{S}_{(w,\infty)}} \mu(d_1) a_{d_1} \mu(d_2) a_{d_2} \Delta^*([d_1, d_2])$$

$$\geq \beta (\frac{\phi(W)}{W} \log R)^{k_0+1}$$

up to errors of $o((\frac{\phi(W)}{W} \log R)^{k_0+1})$ (henceforth referred to as *negligible errors*).

As in Section 2 of the previous post, we can write the left-hand side as

$$\sum_{d_0 \in \mathcal{S}_{(w,\infty)}} \frac{h(d_0)}{d_0} (\sum_{m \in \mathcal{S}_{(w,\infty)}: (m,d_0)=1; m d_0 \in \mathcal{X}} \frac{f'(\frac{\log d_0 m}{\log R})}{\phi(m)})^2 \tag{9}$$

where h is the $k_0 - 1$-dimensional multiplicative function

$$h(d) := \prod_{p \mid d} (k_0 - 1) \frac{(p-1)^2}{p(p-k_0)}.$$

So we would like to select $\mathcal{X}$ small enough that Claim (1) holds, but large enough that we can lower bound (9) by $\beta(\frac{\phi(W)}{W} \log R)^{k_0+1}$ up to negligible errors.

Pintz's idea is to choose $\mathcal{X}$ to be the set of all elements d of $\mathcal{S}_{(w,x^{\delta'})}$ with the property that

$$\prod_{p \mid d: p < x^\delta} p \geq x^{(\delta'-\delta)/2+\varpi+\epsilon/2}. \tag{10}$$

Let us first verify Claim 1 with this definition. Suppose that a_{d_1}, a_{d_2} is non-zero, then from (8) and the support of f there are multiples $d_1 r_1, d_2 r_2$ of d_1, d_2 respectively with $d_1 r_1, d_2 r_2 \in \mathcal{X}$ (in particular $d_1, d_2, r_1, r_2 \in \mathcal{S}_{(w,x^{\delta'})}$) and $d_1 r_1, d_2 r_2 \leq x^{1/4+\varpi}$. Since $d_1 r_1 \in \mathcal{X}$, we have

$$\prod_{p \mid d_1 r_1: p < x^\delta} p \geq x^{(\delta'-\delta)/2+\varpi+\epsilon/2}$$

and thus

$$r_1 \prod_{p \mid d_1: p < x^\delta} p \geq x^{(\delta'-\delta)/2+\varpi+\epsilon/2}.$$

Similarly for r_2 and d_2. Multiplying, we obtain

$$r_1 r_2 (d_1, d_2) \prod_{p \mid [d_1,d_2]: p < x^\delta} p \geq x^{\delta'-\delta+2\varpi+\epsilon}.$$

On the other hand,

$$r_1 r_2 d_1 d_2 \leq x^{1/4+\varpi} x^{1/4+\varpi}$$

and so on dividing we have

$$\frac{1}{[d_1,d_2]} \prod_{p \mid [d_1,d_2]: p < x^\delta} p \geq x^{-1/2+\delta'-\delta+\epsilon}.$$

We conclude that either $q := [d_1,d_2]$ is less than $x^{1/2-\epsilon}$, or else

$$\prod_{p \mid [d_1,d_2]: p < x^\delta} p \geq x^{\delta'-\delta}. \tag{11}$$

The latter conclusion implies that q is x^δ-densely divisible. Indeed, for any $1 \leq R \leq q$ we multiply together all the prime divisors of q between x^δ and $x^{\delta'}$ one at a time until just before one reaches or exceeds R, or if one runs out of such prime divisors. In the former case one ends up at least as large as $x^{-\delta'} R$, and in the latter case one has reached $q / \prod_{p \mid q: p < x^\delta} p$. Next, one multiplies to this the prime divisors of q less than x^δ until one reaches or exceeds $x^{-\delta} R$; this is possible thanks to (11) in the former case and is automatic in the latter case, and gives a divisor between $x^{-\delta} R$ and R as required.

Now we need to obtain a lower bound for (9). If we write

$$F(d_0) := \sum_{m \in \mathcal{S}_{(w,\infty)}: (m,d_0)=1} \frac{-f'(\frac{\log d_0 m}{\log R})}{\phi(m)}$$

and

$$\tilde{F}(d_0) := \sum_{m \in \mathcal{S}_{(w,\infty)}: (m,d_0)=1; m d_0 \in \mathcal{X}} \frac{-f'(\frac{\log d_0 m}{\log R})}{\phi(m)}$$

(the minus sign being to compensate for the non-positive nature of f') then we have

$$0 \leq \tilde{F}(d_0) \leq F(d_0)$$

and thus

$$\tilde{F}(d_0)^2 \geq F(d_0)^2 - 2F(d_0)(F(d_0) - \tilde{F}(d_0)).$$

Note that this inequality replaces the quadratic expression $\tilde{F}(d_0)^2$ with a linear expression in the truncation error $F(d_0) - \tilde{F}(d_0)$, which will be more tractable for computing the effect of that error. We may thus lower bound (9) by the difference of

$$\sum_{d_0 \in \mathcal{S}_{(w,\infty)}} \frac{h(d_0)}{d_0} F(d_0)^2 \tag{12}$$

and

$$2 \sum_{d_0 \in \mathcal{S}_{(w,\infty)}} \frac{h(d_0)}{d_0} F(d_0)(F(d_0) - \tilde{F}(d_0)). \tag{13}$$

In Section 2 of this post it is shown that

$$F(d_0) = (\frac{\phi(W)}{W}\log R)(f(\frac{\log d_0}{\log R}) + O(\frac{d_0}{\phi(d_0)} - 1) + o(1)),$$

which implies that (12) is equal to

$$(\frac{\phi(W)}{W}\log R)^{k_0+1}\int_0^1 f(t)^2\frac{t^{k_0-2}}{(k_0-2)!}\,dt$$

up to negligible errors. Similar considerations show that (13) is equal to

$$2(\frac{\phi(W)}{W}\log R)\sum_{d_0\in\mathcal{S}_{(w,\infty)}}\frac{h(d_0)}{d_0}f(\frac{\log d_0}{\log R})(F(d_0)-\tilde{F}(d_0)) \tag{14}$$

up to negligible errors. To upper bound (14), we need to upper bound

$$F(d_0)-\tilde{F}(d_0) = \sum_{m\in\mathcal{S}_{(w,\infty)}:(m,d_0)=1;md_0\notin\mathcal{X}}\frac{-f'(\frac{\log d_0 m}{\log R})}{\phi(m)}.$$

For this we need to catalog the ways in which md_0 can fail to be in $\mathcal{X}$. In order for this to occur, at least one of the following three statements must hold:

- (i) m could be divisible by a prime p with $x^{\delta'} \le x \le R$.
- (ii) d_0 could be divisible by a prime p with $x^{\delta'} \le x \le R$.
- (iii) d_0 lies in $\mathcal{S}_{(w,x^{\delta'})}$, but $\prod_{p\mid d_0:p<x^\delta} p < x^{(\delta'-\delta)/2+\varpi+\epsilon/2}$.

We consider the contributions of (i), (ii), (iii) to (14). We begin with the contribution of (i). This is bounded above by

$$2(\frac{\phi(W)}{W}\log R)\sum_{x^{\delta'}\le p\le R}\sum_{d_0\in\mathcal{S}_{(w,\infty)}}\frac{h(d_0)}{d_0}$$

$$\sum_{m\in\mathcal{S}_{(w,\infty)}}\frac{-f'(\frac{\log d_0 pm}{\log R})}{\phi(m)\phi(p)}.$$

Applying Lemma 7, one can simplify this modulo negligible errors as

$$2(\frac{\phi(W)}{W}\log R)^2\sum_{x^{\delta'}\le p\le R}\frac{1}{\phi(p)}\sum_{d_0\in\mathcal{S}_{(w,\infty)}}\frac{h(d_0)}{d_0}f(\frac{\log d_0}{\log R})f(\frac{\log d_0}{\log R}+\frac{\log p}{\log R})$$

which by another application of Lemma 7 is equal to

$$2(\frac{\phi(W)}{W}\log R)^{k_0+1}\sum_{x^{\delta'}\le p\le R}\frac{1}{\phi(p)}G_{k_0-1}(0,\frac{\log p}{\log R})$$

where we adopt the notation

$$G_{k_0-1}(t_1,t_2) := \int_0^1 f(t+t_1)f(t+t_2)\frac{t^{k_0-2}}{(k_0-2)!}\,dt.$$

Applying Mertens' theorem and summation by parts, this expression is equal up to negligible errors to

$$2(\frac{\phi(W)}{W}\log R)^{k_0+1}\int_\theta^1 G_{k_0-1}(0,t)\,\frac{dt}{t}$$

where

$$\theta := \frac{\log x^{\delta'}}{\log R} = \frac{\delta'}{1/4+\varpi}.$$

Now we turn to the contribution of (ii) to (14). This is bounded above by

$$2(\frac{\phi(W)}{W}\log R)\sum_{x^{\delta'}\le p\le R}\frac{h(p)}{p}\sum_{d_0\in\mathcal{S}_{(w,\infty)}}\frac{h(d_0)}{d_0}f(\frac{\log pd_0}{\log R})F(pd_0).$$

By Lemma 7 we have

$$F(pd_0)\le(\frac{\phi(W)}{W}\log R)(f(\frac{\log pd_0}{\log R})+o(1))$$

and so we may bound this contribution up to negligible errors by

$$2(\frac{\phi(W)}{W}\log R)^2\sum_{x^{\delta'}\le p\le R}\frac{h(p)}{p}\sum_{d_0\in\mathcal{S}_{(w,\infty)}}\frac{h(d_0)}{d_0}f(\frac{\log pd_0}{\log R})^2$$

which by Lemma 7 again is equal (up to negligible errors) to

$$2(\frac{\phi(W)}{W}\log R)^{k_0+1}\sum_{x^{\delta'}\le p\le R}\frac{h(p)}{p}G_{k_0-1}(\frac{\log p}{\log R},\frac{\log p}{\log R}).$$

By definition, $h(p)=k_0-1+o(1)$. By Mertens' theorem, we can thus write the above expression up to negligible errors as

$$2(k_0-1)(\frac{\phi(W)}{W}\log R)^{k_0+1}\int_\theta^1 G_{k_0-1}(t,t)\,\frac{dt}{t}.$$

Finally, we turn to the contribution of case (iii) to (14). By Lemma 7 we have

$$F(d_0)\le(\frac{\phi(W)}{W}\log R)(f(\frac{\log d_0}{\log R})+o(1))$$

so we may bound this contribution up to negligible errors by

$$2(\frac{\phi(W)}{W}\log R)^2\sum_{d_0}\frac{h(d_0)}{d_0}f(\frac{\log d_0}{\log R})^2$$

where d_0 is as in case (iii).

We introduce the quantities

$$\tilde{\theta} := \frac{\log x^{(\delta'-\delta)/2+\varpi+\epsilon/2}}{\log R} = \frac{(\delta'-\delta)/2+\varpi+\epsilon/2}{1/4+\varpi}$$

and

$$\tilde{\delta} := \frac{\log x^{\delta}}{\log R} = \frac{\delta}{1/4+\varpi}$$

so that case (iii) consists of those d_0 in $\mathcal{S}_{(w,R^\theta)}$ such that

$$\prod_{p \mid d_0 : p < R^{\tilde{\delta}}} p < R^{\tilde{\theta}}.$$

From the support of F we may also take $d_0 \leq R$. This implies that we may factor

$$d_0 = p_1 ... p_J d$$

for some primes

$$R^{\tilde{\delta}} \leq p_1 < ... < p_J \leq R^{\theta}$$

and some $d \leq R^{\tilde{\theta}}$ coprime to $p_1, ..., p_J$, with

$$0 \leq J \leq \frac{1}{\tilde{\delta}}.$$

The contribution of this case may thus be bounded by

$$2(\frac{\phi(W)}{W}\log R)^2 \sum_{0 \leq J \leq \frac{1}{\tilde{\delta}}} \sum_{R^{\tilde{\delta}} \leq p_1 < ... < p_J \leq R^{\theta}} \frac{h(p_1...p_J)}{p_1...p_J}$$

$$\sum_{d \leq R^{\tilde{\theta}}} \frac{h(d)}{d} f(\frac{\log dp_1...p_J}{\log R})^2.$$

Evaluating the inner sum using Lemma 7, we obtain (up to negligible errors)

$$2(\frac{\phi(W)}{W}\log R)^{k_0+1} \sum_{0 \leq J \leq \frac{1}{\tilde{\delta}}} \sum_{R^{\tilde{\delta}} \leq p_1 < ... < p_J \leq R^{\theta}} \frac{h(p_1...p_J)}{p_1...p_J}$$

$$G_{k_0-1,\tilde{\theta}}(\frac{\log p_1...p_J}{\log R}, \frac{\log p_1...p_J}{\log R})$$

where $G_{k_0-1,\tilde{\theta}}$ is a truncation of G_{k_0-1}:

$$G_{k_0-1,\tilde{\theta}}(t_1,t_2) := \int_0^{\tilde{\theta}} f(t+t_1)f(t+t_2)\frac{t^{k_0-2}}{(k_0-2)!}\,dt.$$

Again we have $h(p_1...p_J) = (k_0-1)^J + o(1)$. By Mertens' theorem we may write this (up to negligible errors) as

$$2(\frac{\phi(W)}{W}\log R)^{k_0+1}\sum_{0\leq J\leq\frac{1}{\tilde{\delta}}}(k_0-1)^J\int_{\tilde{\delta}\leq t_1<\ldots<t_J\leq\theta}$$

$$G_{k_0-1,\tilde{\theta}}(t_1+\ldots+t_J,t_1+\ldots+t_J)\,\frac{dt_1\ldots dt_J}{t_1\ldots t_J}.$$

Putting all this together, we have obtained the lower bound (6) with

$$\beta=G_{k_0-1}(0,0)(1-2\kappa_1-2\kappa_2-2\kappa_3)$$

where

$$\kappa_1:=G_{k_0-1}(0,0)^{-1}\int_\theta^1 G_{k_0-1}(0,t)\,\frac{dt}{t}$$

$$\kappa_2:=(k_0-1)G_{k_0-1}(0,0)^{-1}\int_\theta^1 G_{k_0-1}(t,t)\,\frac{dt}{t}$$

and

$$\kappa_3=G_{k_0-1}(0,0)^{-1}\sum_{0\leq J\leq\frac{1}{\tilde{\delta}}}(k_0-1)^J\int_{\tilde{\delta}\leq t_1<\ldots<t_J\leq\theta}$$

$$G_{k_0-1,\tilde{\theta}}(t_1+\ldots+t_J,t_1+\ldots+t_J)\,\frac{dt_1\ldots dt_J}{t_1\ldots t_J}.$$

We now place upper bounds on $\kappa_1,\kappa_2,\kappa_3$. In this previous post, the bounds

$$G_{k_0-1}(0,t)\leq(1-t)^{(k_0-1)/2}G_{k_0-1}(0,0)$$

and

$$G_{k_0-1}(t,t)\leq(1-t)^{k_0-1}G_{k_0-1}(0,0)$$

for $0<t<1$ are proven. Thus we have

$$\kappa_1\leq\int_\theta^1(1-t)^{(k_0-1)/2}\,\frac{dt}{t}$$

$$\kappa_2\leq(k_0-1)\int_\theta^1(1-t)^{k_0-1}\,\frac{dt}{t}.$$

These are already quite small for $\theta\approx 1/2$, say, which would correspond to $\delta'\approx 1/8$.

For κ_3 we will use the crude estimate

$$G_{k_0-1,\tilde{\theta}}(t_1+\ldots+t_J,t_1+\ldots+t_J)\leq G_{k_0-1,\tilde{\theta}}(0,0);$$

this may surely be improved, but we will not do so here to simplify the exposition. Then we may bound

$$\kappa_3 \leq \frac{G_{k_0-1,\tilde\theta}(0,0)}{G_{k_0-1}(0,0)} \sum_{0\leq J\leq 1/\tilde\delta} \frac{(k_0-1)^J}{J!} (\log \frac{\theta}{\tilde\delta})^J.$$

The point here is that the first term $\frac{G_{k_0-1,\tilde\theta}(0,0)}{G_{k_0-1}(0,0)}$ is exponentially decaying in k_0, which can compensate for the second term if $1/\tilde\delta \ll k_0$ which is currently the case in the regime of interest.

One can do a bit better than this. For any parameter $A \geq 0$, one has

$$G_{k_0-1,\tilde\theta}(t_1+\ldots+t_J, t_1+\ldots+t_J) \leq e^A e^{-A(t_1+\ldots+t_J)} G_{k_0-1,\tilde\theta}(0,0)$$

since the left-hand side vanishes for $t_1+\ldots+t_J \geq 1$. This gives the bound

$$\kappa_3 \leq e^A \frac{G_{k_0-1,\tilde\theta}(0,0)}{G_{k_0-1}(0,0)} \sum_{0\leq J\leq 1/\tilde\delta} \frac{(k_0-1)^J}{J!} (\int_{\tilde\delta}^{\theta} e^{-At} \frac{dt}{t})^J.$$

If we insert these bounds into (7), send ϵ to zero, and optimise in f using Theorem 14 from this previous post, we obtain Theorem 5.

Comments (82)

Andrew Sutherland · 18 June 2013

Has anyone worked out the "cheap" gain k0 mentioned after Theorem 6? (or better, a fully optimized k0, but even a cheap gain will give those working on optimizing H(k0) something to play with — we're not used to working with the same k0 for more than 2 days at time :)).

v08ltu · 18 June 2013

If I typed it all correctly, $\omega = 2/733$ allows $k_0 = 5937$.

```
Bz(k)=k+1.85575708149*k^(1/3)+1.033150/k^(1/3);
 DELTA(om)=(1/4-87*om)/17;
 KAPPA1(om,k0)=local(d);d=DELTA(om);intnum(t=4*d/(1+4*om),1,(1-t)^((k0-1)/2)/t);
 KAPPA2(om,k0)=local(d);\
 d=DELTA(om);(k0-1)*intnum(t=4*d/(1+4*om),1,(1-t)^((k0-1))/t);

INEQ(om,k0)=(1+4*om)*(1-2*KAPPA1(om,k0)-2*KAPPA2(om,k0))-Bz(k0-2)^2/k0/(k0-1);

INEQ(2/733,5937)
```

Andrew Sutherland · 18 June 2013

Thanks!

v08ltu · 18 June 2013

If I use the above κ_3 bound and assume the G-quotient really is $\tilde\theta^{k0-1}$, I think that I get that $\omega = 1/348 - 151/2000000$ and $\delta' = \delta + 1/100$ with δ to satisfy the previous constraint, one can get $k_0 = 5672$. I should write a better program to optimize this though.

Terence Tao · 18 June 2013

From my other comment, I now think the G-quotient is going to be closer to $\exp(-k_0(\alpha - \tanh\,\alpha + o(1)))$, where $\alpha := \mathrm{sech}^{-1}\tilde\theta$. I can try to get more precise

asymptotics for the $o(1)$ term, but perhaps it will be simpler just to compute the integral numerically (it will serve as a good sanity check to compare the numerics against the asymptotics though, and the asymptotics can give a rough idea of what region of parameter space to search in).

v08ltu · 18 June 2013

In the range of interest, I seem to have $\kappa_3 \approx 10^{-4922}$ and the other κ are also 10^{-54} and 10^{-102} respectively. So none of them matter. This assumes I am not making any errors or typos with formulas.

v08ltu · 18 June 2013

I am really not sure where I came up with the above numbers, there is no obstacle to making ω closer to 1/348. With the sech/tanh estimate (ignoring o(1)) I can take $\omega = 1/348 - 9/2000000$ and get $k_0 = 5459$ with $\delta' - \delta = 1/130$, and κ_3 is not of import again (make the delta-difference smaller if it is). I am not sure I trust all the numerics though (instability), but this is approaching the limit of previous analysis.

v08ltu · 18 June 2013

Looks like $k - 0 = 5456$ with $\omega = 1/348 - 7/2000000$ and $\delta' - \delta = 1/300$ works. If you try to take ω smaller, then κ_1 becomes not negligible, and you have the balance the delta-difference. Too hard to look at all parameters right now.

Ooops, looks like $\omega = 1/348 - 5/2000000$ and $\delta' - \delta = 1/800$ and $k_0 = 5454$ works (with all κ in play).

v08ltu · 18 June 2013

You can also make $\delta' - \delta = 1/900$ in this last one, and then κ_3 is irrelevant (10^{-89} in sech/tanh), while κ_1 is then slightly bigger (5x), but not enough to vitiate key inequality.

v08ltu · 18 June 2013

Managed to squeeze one more, $\omega = 1/348 - 5.1/2000000$ (larger than previous!) with a δ-difference of (say) $1/750$ allows $k_0 = 5453$.

v08ltu · 18 June 2013

Slightly nicer form is $k_0 = 5453$ with $\omega = 1/348 - 1/400000$ and $\delta' - \delta \in [1/850, 1/810]$. I never checked that making δ smaller than forced by ω would not gain, but I don't see why it would (probably follows by monotonicity).



Terence Tao · 19 June 2013

Cross-checking these computations with Maple. With $k_0 = 5453$ and $\varpi = 1/348 - 1/400000$ we have $\delta = 87/6800000$. Setting $\delta' := 1/820$ (say) we have

$\theta = 1740000/360796433$

$\tilde{\theta} = 98909032/6133539361$

$\tilde{\delta} = 7569/149598521.$

We need $2\kappa_1 + 2\kappa_2 + 2\kappa_3$ to not exceed

$1 - \frac{j_{k_0-2}^2}{k_0(k_0-1)(1+4\varpi)} = 0.3883 \times 10^{-6}$.

I get

$\kappa_1 = 0.1336 \times 10^{-6}$

$\kappa_2 = 0.7118 \times 10^{-9}$

which looks good so far. Now for κ_3. Using the crude bound

$\kappa_3 \leq \frac{G_{k_0-1,\tilde\theta}(0,0)}{G_{k_0-1}(0,0)} \exp(2(k_0-1)^{1/2}\tilde\theta^{-1/2})$

the exponential factor here is 0.111×10^{506}, which looks scary. On the other hand, the G-ratio here is predicted to be roughly

$\exp(-2k_0(\alpha - \tanh(\alpha))) \approx 0.7615 \times 10^{-8340}$

with $\mathrm{sech}(\alpha) = \tilde\theta^{1/2}$ (I made a slight mistake earlier in not having the square root and factor of 2 here but it fortunately turns out not to matter), and this is is completely negligible. In order to get an independent confirmation of the smallness of the G-ratio is, the Bessel function J_{k_0-2} at $\sqrt{\tilde\theta} j_{k_0-2}$ takes the value 0.1805×10^{-4156}, and at $0.999 j_{k_0-2}$ takes the value 0.0187, so it looks abundantly clear that the G-ratio is indeed ridiculously small in this regime.



v08ltu · 19 June 2013

You have an error in your κ_3 expression in the Maple check, it is $\tilde\delta$ in the denominator of the sqrt inside the exp, not $\tilde\theta$. I think the sech-sqrt fix breaks $k_0 = 5453$, I will check.



Terence Tao · 19 June 2013

Oops, sorry about that. Well, one can hope to squeeze something out of the full version of Theorem 5 (or at least the first display after Theorem 5) in which there is also the A parameter to optimise over. One can probably pick up some other ways to make the estimate for κ_3 more efficient, but at the cost of simplicity. Pintz's own notes have a slightly different way of estimating κ_3, I can try to have a look at that instead.

There were two corrections I had to make; one (which worked against our favour) was the replacing of $\tilde\theta$ with its square root, but the other (which works for us) is the doubling of the exponential from $\exp(-k_0(\alpha - \tanh(\alpha)))$ to $\exp(-2k_0(\alpha - \tanh(\alpha)))$. I had thought that the two effects roughly cancelled each other out, but I guess not quite...



v08ltu · 19 June 2013

Currently, I can only get $k_0 = 5455$ with $\omega = 1/348 - 6/2000000$ and $\delta' \in [1/720, 1/900]$. You can note the incredible impact that κ_3 suddenly has if you change "6" to "5" in the numerator of the subtrahend of ω. So saving a few more in k_0 could occur from its more benevolent estimation.



Terence Tao · 19 June 2013

I just realised that the estimate

$$\int_{\tilde\delta}^{\theta} e^{-At}\frac{dt}{t} \leq \frac{1}{A\tilde\delta}$$

that I used below Theorem 6 can be improved to

$$\int_{\tilde\delta}^{\theta} e^{-At}\frac{dt}{t} \leq \frac{e^{-A\tilde\delta}}{A\tilde\delta}$$

(by replacing θ with infinity and replacing $\frac{1}{t}$ with $\frac{1}{\tilde\delta}$). This extra exponential factor makes it harder to optimise in A, but it should give a significant savings, hopefully enough to push back to 5453 and maybe even a bit further.



Terence Tao · 18 June 2013

I'm recording here some asymptotics of Bessel functions to help understand the ratio

$$\frac{G_{k_0-1,\tilde\theta}(0,0)}{G_{k_0-1}(0,0)} = \frac{\int_0^{\tilde\theta^{1/2} j_{k_0-2}} J_{k_0-2}(x)^2\ x dx}{\int_0^{j_{k_0-2}} J_{k_0-2}(x)^2\ x dx}\ (*)$$

in the regime where $0 < \tilde\theta < 1$ is fixed and k_0 is large. Here we have made the change of variable $t = (x/j_{k_0-2})^2$.

Write $\nu := k_0 - 2$. By 9.3.23 of Abramowitz and Stegun (see e.g. this link) we have the asymptotic

$$J_\nu(\nu + z\nu^{1/3}) = \frac{2^{1/3}}{\nu^{1/3}}\ \mathrm{Ai}(-2^{1/3}z)(1+o(1))$$

for bounded z, where Ai is the Airy function. This shows that the zero j_ν has the asymptotic

$$j_\nu = \nu + 2^{-1/3}c\nu^{1/3} + o(\nu^{1/3})$$

where $-c$ is the first negative zero of the Airy function. It also suggests that

$$\int_0^{j_{k_0-2}} J_{k_0-2}(t)^2\ t dt \approx 2^{-2/3}\nu^{-1/3}\int_{-c}^{\infty} \mathrm{Ai}(z)^2\ dz.$$

Meanwhile, 9.3.7 of Abramowitz and Stegun (Debye asymptotic) tells us that

$$J_\nu(\nu\ \mathrm{sech}\ \alpha) \approx e^{\nu(\tanh\alpha - \alpha + o(1))}$$

for any $\alpha > 0$ (this is also consistent with the exponential decay of the Airy function to the right), so if we set $\alpha := \mathrm{sech}^{-1}\tilde\theta$ (EDIT: this should be $\alpha = \mathrm{sech}^{-1}\tilde\theta^{1/2}$), the ratio (*) should take the form

$$e^{\nu(\tanh\alpha - \alpha + o(1))}$$

(EDIT: this should be $e^{2\nu(\tanh\alpha - \alpha + o(1))}$) which is exponentially decaying in k_0 as claimed in the post. (Unfortunately, we cannot directly use these asymptotics to get explicit values of k_0 because we need control on the error terms, which standard sources (e.g. Abramowitz-Stegun) do not supply.)



Eytan Paldi · 19 June 2013

Please note that the upper bound for the integration in the numerator is defined in theorem 5 to be $\tilde\theta$ (not its square), and in the presentation of Pintz argument it is defined

(for a general pair
(t_1, t_2)) to be $\tilde{\delta}$, so it is not clear which is the correct one.

[Corrected, thanks -T.]

Eytan Paldi · 19 June 2013

In my last message I did not notice that the integrals are after change of variable, but I think that in the numerator $\tilde{\theta}$ should be replaced by its square root. And more importantly, the integral (at least for even k_0) can be precisely evaluated as a finite linear combination of squares of Bessel functions by finding the indefinite integral using the recursive formula (11.3.33) and the formula (11.3.34) in Abramowitz and Stegun.
(If there is a corresponding formula for $k_0 - 2 = 1$, then the recursive formula can be used to compute the integral for odd k_0 as well.)

coco · 19 June 2013

typo: divsibility -> divisibility, in second paragraph.

[Corrected, thanks – T.]

Gergely Harcos · 19 June 2013

The nonnegativity property I tried to prove in the previous thread (https://terrytao.wordpress.com/2013/06/11/further-analysis-of-the-truncated-gpy-sieve/) is equivalent to

$\langle \alpha_0, \alpha_0 \rangle_{t_1,\dots,t_n} \geq 0$ for any $0 \leq t_1, \dots, t_n < 1$.

Unfortunately, SAGE found the following counterexample for $k_0 = 2$ and $n = 5$:

$\{t_1, t_2, t_3, t_4, t_5\} = \{0.20, 0.21, 0.24, 0.25, 0.31\}$.

More precisely, with these parameters we have

$\sum_{\{1,\dots,5\}=S\cup T} (-1)^{|S\cap T|} \min((1-\sum_{i\in S} t_i)_+^2, (1-\sum_{i\in T} t_i)_+^2) = -0.0824.$

This result is verified by the SAGE commands

A={20/100, 21/100, 24/100, 25/100, 31/100}

print sum([((-1)^U.cardinality())*sum([min(max(0,1-sum(S.union(U))),max(0,1-sum(A.difference(S))))^2 for S in Subsets(A.difference(U))]) for U in Subsets(A)])

Terence Tao · 19 June 2013

A minor technical point: there is one tiny way in which x^δ-densely divisible numbers are worse than x^δ-smooth numbers: the behaviour is not quite hereditary with respect to factors. If $d = d_1 d_2$ and d is x^δ-smooth, then d_1 is automatically x^δ-smooth as well. On the other hand, if d is merely x^δ-densely divisible, then d_1 is only $x^\delta d_2$-densely divisible, there is a loss of d_2 (basically because d_2 could suck up all the useful small primes that are giving the dense divisibility property). This makes the Type III analysis a little more complicated as there is a part in which one forces the frequency parameter h to be coprime to the modulus d by Mobius inversion, dividing both h and d by the common factor (h, d). However even with the worsening of x^δ by (h, d), one can check that the net power of (h, d) remains favorable and the coprime case still dominates, so this is not actually an issue. (But something to look out for when we do the final writing up of the arguments.)

Terence Tao · 19 June 2013

For reference, I have uploaded (with permission) Pintz's original notes on this sieve here:

Section 1

Section 2

Section 3

v08ltu · 19 June 2013

OK, I think with $A = 9500$ and $\omega = 1/348 - 5/2000000$ and $\delta' = 1/800$ this now recovers $k_0 = 5453$.

v08ltu · 19 June 2013

And $\omega = 1/348 - 4/2000000$ with $\delta' = 1/820$ seems to give $k_0 = 5452$ with the same $A = 9500$.

Terence Tao · 20 June 2013

I verified that this works with the Maple code below (using crude upper and lower bounds on the Bessel integral after verifying that the Bessel function J_{k_0-2} was decreasing from $0.998j_{k_0-2}$ to $0.999j_{k_0-2}$, and was on the other side of the maximum at $\sqrt{\tilde{\theta}}j_{k_0-2}$). By using the κ_3 bound in the display after Theorem 6 rather than perform any further estimations, there is actually a little bit of margin: $2\kappa_1 + 2\kappa_2 + 2\kappa_3$ is allowed to be as large as 0.100×10^{-5} but is only of size 0.269×10^{-6}, with κ_1 dominating, $\kappa_2 \approx 0.715 \times 10^{-9}$, and $\kappa_3 \leq 10^{-2454}$. It seems likely that we could squeeze k_0 by one or two more by changing the free parameters ϖ, δ', A a bit, but it is very tight and I didn't try too hard to do so.

```
varpi := 1/348 - 4/2000000;
k0 := 5452;
deltap := 1/820;
A := 9500;

delta := (1/4 - 87 * varpi) / 17;
theta := deltap / (1/4 + varpi);
thetat := (deltap - delta + varpi) / (1/4 + varpi);
deltat := delta / (1/4 + varpi);

j := BesselJZeros(k0-2,1);

eps := 1 - j^2 / (k0 * (k0-1) * (1+4*varpi));

kappa1 := int( (1-t)^((k0-1)/2)/t, t = theta..1, numeric);
kappa2 := (k0-1) * int( (1-t)^(k0-1)/t, t=theta..1, numeric);

e := exp( A + (k0-1) * int( exp(-A*t)/t, t=deltat..theta, numeric ) );

s1 := BesselJ(k0-2,0.999*j);
s2 := BesselJ(k0-2,0.998*j);
s3 := BesselJ(k0-2,0.997*j);
```

check that s3 > s2 > s1 to ensure Bessel function decreases from 0.998 j to 0.999 j

crude lower bound on the denominator integral
gd := s1^2 * 0.001 * j * 0.998 * j;

crude upper bound on the numerator integral
tn := sqrt(thetat)*j;
s0 := BesselJ(k0-2,tn);
gn := s0^2 * tn * tn / 2;

check that s0 < s3 to ensure that Bessel function decreases up to tn (but this is true with massive amounts of room to spare)

kappa3 := (gn/gd) * e;

eps2 := 2*(kappa1+kappa2+kappa3);

we win if eps2 < eps

abqmathteacher · 20 June 2013

A quick nontechnical question. Any idea of the likelihood of pushing this line of reasoning down to H < 1476, the largest known (I think) "maximal prime gap" (see http://oeis.org/A005250, http://oeis.org/A005250/a005250.txt)? That would be an exciting landmark.

Terence Tao · 20 June 2013

Actually it is known that prime gaps can be arbitrarily large. For instance, if n is a natural number, then $n! + 2, n! + 3, \ldots, n! + n$ are all composite, creating a prime gap of length at least n. I think 1476 may be the smallest H for which the location of the first prime gap of length H is known, though.

My feeling though is that we are going to run out of improvements before we reach that level; $H = 10^5$ looks like a more realistic target at this stage.

abqmathteacher · 20 June 2013

Thanks! Indeed the term “maximal prime gap” used in the OEIS entry is confusing, sorry to just quote it. It means “maximal gap for a given size integer”. So 1476 is (according to that table) the *largest* H for which the location of the first prime gap of length H is known.
Anyway, even H = 10^5 is quite exciting.
David Metzler

Dave · 20 June 2013

If I’m reading the polymath wiki correctly, isn’t the best known H value already below 10^5?

Terence Tao · 20 June 2013

Er, sorry, I meant $H = 10^4$ would be a reasonable end target for this project. :)

Gergely Harcos · 20 June 2013

1. If j'_{k_0-2} denotes the first zero of $J'_{k_0-2}(x)$, then it is known that

$$k_0 - 2 < j'_{k_0-2} < j_{k_0-2} < \sqrt{2}k_0,$$

see Section 15.3 in Watson: A treatise on the theory of Bessel functions (2nd ed., Cambridge University Press, 1944). Hence for $\tilde{\theta} < \frac{1}{3}$ and $k_0 \geq 11$ we can bound

$\int_0^{\tilde{\theta}^{1/2} j_{k_0-2}} J_{k_0-2}^2(x)\ x dx < \frac{1}{2}\tilde{\theta} j_{k_0-2}^2 J_{k_0-2}^2(\tilde{\theta}^{1/2} j_{k_0-2})$.

On the other hand, by an orthogonality relation of Bessel functions (see Wikipedia)

$\int_0^{j_{k_0-2}} J_{k_0-2}^2(x)\ x dx = j_{k_0-2}^2 \int_0^1 J_{k_0-2}^2(j_{k_0-2} t)\ t dt$

$= \frac{1}{2} j_{k_0-2}^2 J_{k_0-1}^2(j_{k_0-2})$.

It follows that

$\frac{\int_0^{\tilde{\theta}^{1/2} j_{k_0-2}} J_{k_0-2}^2(x)\ x dx}{\int_0^{j_{k_0-2}} J_{k_0-2}^2(x)\ x dx} < \tilde{\theta} \frac{J_{k_0-2}^2(\tilde{\theta}^{1/2} j_{k_0-2})}{J_{k_0-1}^2(j_{k_0-2})}$.

SAGE tells me that for $k_0 = 5452$ and $\tilde{\theta} = 98909032/6133539361$ the right hand side is less than 1.3836×10^{-8309}.

2. Some misprints: "analyhsis", "systetm".

[Corrected, thanks. These calculations clean up the rather ad hoc Maple code I had previously! -T]

Eytan Paldi · 20 June 2013

Please note that if k_0 is even , than by formulas (11.3.33) and (11.3.34) in Abramowitz & Stegun both integrals can be expressed (and thereby precisely computed) as a simple finite linear combinations of squares of Bessel functions! (For odd k, one needs an analogous formula to (11.3.34) in which $J_1(t)$ replaces $J_0(t)$.)

Gergely Harcos · 20 June 2013

Thank you! Actually this works for J_ν with any $\nu \geq 0$, see my latest comment below.

Gergely Harcos · 20 June 2013

Actually we can calculate the ratio

$\frac{\int_0^{\tilde{\theta}^{1/2} j_{k_0-2}} J_{k_0-2}^2(x)\ x dx}{\int_0^{j_{k_0-2}} J_{k_0-2}^2(x)\ x dx}$

explicitly in terms of Bessel values. According to Lemma 5.4 in Folland: Fourier analysis and its applications,

$\int_0^b J_\nu^2(x)\ x dx = \frac{b^2}{2} J_\nu'^2(b) + \frac{b^2-\nu^2}{2} J_\nu^2(b)$ for any $b > 0$ and $\nu \geq 0$,

where we can express $J_\nu'(x)$ as $J_{\nu-1}(x) - J_{\nu+1}(x)$. Using this, SAGE now tells me (for $k_0 = 5452$ and $\tilde{\theta} = 98909032/6133539361$) that the numerator is $\approx 4.9217 \times 10^{-8311}$, and the denominator is $\approx 1.9231 \times 10^2$, so that

$\frac{\int_0^{\tilde{\theta}^{1/2} j_{k_0-2}} J_{k_0-2}^2(x)\ x dx}{\int_0^{j_{k_0-2}} J_{k_0-2}^2(x)\ x dx} \approx 2.5592 \times 10^{-8313}$.

Gergely Harcos · 20 June 2013

Small correction: "$J_\nu'(x)$ as $J_{\nu-1}(x) - J_{\nu+1}(x)$" should be "$J_\nu'(b)$ as $\frac{J_{\nu-1}(b) - J_{\nu+1}(b)}{2}$".

Eytan Paldi · 20 June 2013

Nice ! (such simple expression is quite surprising!)

Eytan Paldi · 20 June 2013

It seems that there is a slightly simpler expression (for integer order n):

$\int_0^x tJ_n^2(t)dt = \frac{x^2}{2}[J_n^2(x) - J_{n-1}(x)J_{n+1}(x)]$

This is formula (8) in Auluck's arxiv paper :

http://arxiv.org/abs/1006.4417

Gergely Harcos · 20 June 2013

Very good. Actually this formula holds for all $n \geq 0$ (not only for integers): it can be derived from Folland's lemma, using $2J'_\nu(x) = J_{\nu-1}(x) - J_{\nu+1}(x)$ and $2\nu J_\nu = x(J_{\nu-1}(x) + J_{\nu+1}(x))$.

Eytan Paldi · 20 June 2013

I think that it even holds (by analytic continuation) for all complex ν (since the integrand is entire as a function of ν for each fixed nonzero t.)

Gergely Harcos · 21 June 2013

Eytan: in extending the formula to all complex ν, one needs to be careful, e.g. one needs to check if the integral converges (hopefully locally uniformly, the difficulty coming from $t \approx 0$). Already the extension of $J_\nu(x)$ is tricky to all complex ν. Certainly everything is fine for $\Re(\nu) > -\frac{1}{2}$.

Eytan Paldi · 21 June 2013

I agree that the integral can't be entire in ν, since for any fixed $x > 0$, (9.1.10) in A&S implies that the integral diverges as
ν tends (from the right) to $\nu = -1$ – which must therefore be a singular point for any analytic continuation.

Gergely Harcos · 21 June 2013

Eytan: Right. Probably there is no problem for $\Re(\nu) > -1$, as the integral converges absolutely and locally uniformly there. I sticked to $\Re(\nu) > -\frac{1}{2}$ just to be safe, because the definition beyond that range is less straightforward (according to Watson's book). BTW I am also quite pleased and surprised that so many transparent features of Bessel functions play a role in this project. I come from automorphic forms where this phenomenon is wide-spread, but (I think) it is new for sieve theory where this project belongs.

Eytan Paldi · 21 June 2013

There is another interesting "sub-multiplicative" property of f_0 (shared also by many entire functions of order < 2) which is very useful for improving the current bounds on the κ's. (I'm preparing the details.)

Exciting real-time developments about prime numbers | Mountain of Math · 21 June 2013

[...] I asked about this at Terry Tao's blog, and he indicated that H = 10,000 would be a reasonable target for the polymath8 project. And in [...]

Eytan Paldi · 22 June 2013

In the sum defining κ_3, the sum variable J is upper bounded by $1/\delta$, but the original range of J (17 lines above) is
$0 \le J \le 1/\tilde{\delta}$. So it is not clear which upper bound for J is the correct one.

[The correct range is $J \le 1/\tilde{\delta}$. I was not able to locate any instance where $1/\delta$ was used instead, although sometimes the tilde is almost obscured by the dividing bar between the numerator and denominator. -T]

Gergely Harcos · 22 June 2013

1. I have difficulty in deriving the fourth display below (8), i.e. that the left hand side of (5) is up to negligible error

$\sum_{d_0 \in \mathcal{X}} \frac{1}{\Phi(d_0)} f'(\frac{\log d_0}{\log R})^2$.

This formula should come from

$\sum_{d_0 \in \mathcal{S}_I} \Phi(d_0)(\sum_{d \in \mathcal{S}_I : d_0 \mid d} \frac{\mu(d)}{\Phi(d)} \sum_{q \in \mathcal{X} : (q,d)=1} \frac{1}{\Phi(q)} f'(\frac{\log dq}{\log R}))^2$,

where $I = (w, +\infty)$. Writing $d = d_0 d_1$ and $m = d_1 q$, the inner sum becomes

$\frac{\mu(d_0)}{\Phi(d_0)} \sum_{m \in \mathcal{S}_I : (m,d_0)=1} \frac{f'(\frac{\log d_0 m}{\log R})}{\Phi(m)} \sum_{d_1 \mid m; m/d_1 \in \mathcal{X}} \mu(d_1)$.

Without the condition $m/d_1 \in \mathcal{X}$ in the innermost sum, we would get in the end

$\sum_{d_0 \in \mathcal{S}_I} \frac{1}{\Phi(d_0)} f'(\frac{\log d_0}{\log R})^2$,

which is admissible. However, I don't see how to handle the condition $m/d_1 \in \mathcal{X}$ (it was automatic for the earlier choices of $\mathcal{X}$).

2. I don't see in (9) the condition $md_0 \in \mathcal{X}$. Certainly $md_0 = dq$ is the product of $d \in \mathcal{S}_I$ and $q \in \mathcal{X}$, but why does it lie in $\mathcal{X}$?

> **Terence Tao** · 22 June 2013
>
> Oops, I think I mis-stated the constraint in (8) which is causing both of these problems: it should be dq that lies in $\mathcal{X}$ rather than q. I'll edit the post accordingly.
>
> > **Gergely Harcos** · 22 June 2013
> >
> > Thank you, it is all clear now. I just posted a small improvement, I hope it is ok.

Gergely Harcos · 22 June 2013

1. I think the display below (11) can be improved to

$\prod_{p \mid q : p < x^\delta} p \ge x^{2\delta' - 2\delta}$.

By assumption, $q = [d_1, d_2]$ such that $d_1 r_1, d_2 r_2 \in \mathcal{X} \cap [1, R]$ for some $r_1, r_2 \in \mathcal{S}_I$.
It follows that

$(r_1 \prod_{p \mid d_1 : p < x^\delta} p)(r_2 \prod_{p \mid d_2 : p < x^\delta} p) \ge x^{2\delta' - 2\delta + 2\varpi + \epsilon}$,

whence

$r_1 r_2 (d_1, d_2)(\prod_{p \mid [d_1,d_2] : p < x^\delta} p) \ge x^{2\delta' - 2\delta + 2\varpi + \epsilon}$.

Multiplying by $[d_1,d_2]R^{-2}$ and using $(d_1r_1)(d_2r_2) \leq R^2$ we obtain

$\prod_{p \mid [d_1,d_2]: p < x^\delta} p \geq x^{2\delta'-2\delta-1/2+\epsilon}[d_1,d_2]$.

Therefore either $[d_1,d_2] \leq x^{1/2-\epsilon}$ or

$\prod_{p \mid q: p < x^\delta} p \geq x^{2\delta'-2\delta}$.

Hopefully this improvement lowers k_0 somewhat.

2. "Section 2 of the previous section" should be "Section 2 of this previous blog".



Terence Tao · 22 June 2013

Thanks for the correction and improved argument! I edited the main post accordingly. The net effect is to improve the quantity $\tilde{\theta}$ in Theorem 5 from

$$\tilde{\theta} := \frac{\delta' - \delta + \varpi}{1/4 + \varpi}$$

to

$$\tilde{\theta} := \frac{(\delta' - \delta)/2 + \varpi}{1/4 + \varpi}$$

which should hopefully squeeze a bit more out of k_0 starting from the current constraint $148\varpi + 33\delta < 1$ since it delays the time when the κ_3 error starts exploding and so δ' may be taken a bit larger than previously. Presumably some perturbation of v08ltu's choice of parameters

$$\varpi = 1/148 - 24/4000000, \delta' = 1/200, k_0 = 1467, A = 1300$$

should work here to nudge k_0 below 1467...



v08ltu · 22 June 2013

For instance, $\omega = 1/148 - 12/4000000, \delta' = 1/180, k_0 = 1466, A = 1200$.



Terence Tao · 22 June 2013

~~Confirmed! H is now 12006.~~ *[Retracted, see below. -T]*

```
varpi := 1/148 - 12/4000000;
k0 := 1466;
deltap := 1/180;
A := 1200;

delta := (1 - 148 * varpi) / 33;
theta := deltap / (1/4 + varpi);

# Gergely's improved value for thetat
thetat := ((deltap - delta)/2 + varpi) / (1/4 + varpi);
deltat := delta / (1/4 + varpi);

j := BesselJZeros(k0-2,1);

eps := 1 - j^2 / (k0 * (k0-1) * (1+4*varpi));
```

```
kappa1 := int( (1-t)^((k0-1)/2)/t, t = theta..1, numeric);
 kappa2 := (k0-1) * int( (1-t)^(k0-1)/t, t=theta..1, numeric);

e := exp( A + (k0-1) * int( exp(-A*t)/t, t=deltat..theta, numeric ) );

# using Gergely's exact expression for denominator
 gd := (j^2/2) * BesselJ(k0-3,j)^2;

# using Eytan's exact expression for numerator
 tn := sqrt(thetat)*j;
 gn := (tn^2/2) * (BesselJ(k0-2,tn)^2 - BesselJ(k0-3,tn)*BesselJ(k0-1,tn));

kappa3 := (gn/gd) * e;

eps2 := 2*(kappa1+kappa2+kappa3);

# we win if eps2 < eps
```

Hannes · 22 June 2013

With that script I get eps2-eps=1.268156732*10^251 (not good).

Hannes · 22 June 2013

(I used Maple 14)

Terence Tao · 22 June 2013

Hmm, you're right! My version of maple seems to be a bit glitchy, but it is now reporting the same answer as you – once again κ_3 is blowing up. So I have to retract the confirmation for now...

v08ltu · 23 June 2013

I found my error, that I divided ω by 2 also in the $\tilde{\theta}$ numerator, not only the δ difference.

v08ltu · 23 June 2013

The main issue that you need to take $\omega = 1/148 - 12/4000000$ or less to have a hope from Bessel zeroes with $k_0 = 1466$, and this is not able to work, as the ω part dominates the $\tilde{\theta}$ numerator (at least currently).

Gergely Harcos · 22 June 2013

Let $q^\flat := \prod_{p \mid q : p < x^\delta} p$ and $q^\sharp := \prod_{p \mid q : p \geq x^\delta} p$, then the paragraph below (11) seems to work only when $q^\sharp \geq R$ (e.g. for $q^\sharp = 1$ the first step gives nothing). For $q^\sharp < R$ we can do the following. We have $q^\flat \geq R/q^\sharp > 1$, hence there is a divisor of $q^\flat$ in the interval $[Rx^{-\delta}/q^\sharp, R/q^\sharp]$. Multiplying this divisor by $q^\sharp$ yields a divisor of q in the interval $[Rx^{-\delta}, R]$.

[Corrected, thanks – T.]

Gergely Harcos · 22 June 2013

1. In the post and in the comments we used the bound (valid for any $A > 0$)

$$\int_{\tilde{\delta} \leq t_1, \dots, t_J \leq \theta; \sum t_j \leq 1} \frac{dt_1 \dots dt_J}{t_1 \dots t_J} < e^A \Big(\int_{\tilde{\delta}}^{\theta} e^{-At} \frac{dt}{t} \Big)^J.$$

Here is an alternate way to bound the left hand side, which probably performs better in practice without any auxiliary parameter:

$\int_{\tilde\delta \leq t_1,\dots,t_J \leq \theta; \sum t_j \leq 1} \frac{dt_1\dots dt_J}{t_1\dots t_J}$

$= \int_{1 \leq \tau_1,\dots,\tau_J \leq \theta/\tilde\delta; \sum \tau_j \leq 1/\tilde\delta} \frac{d\tau_1\dots d\tau_J}{\tau_1\dots\tau_J}$

$\leq \int_{1 \leq \tau_1,\dots,\tau_J; \sum \tau_j \leq 1/\tilde\delta} d\tau_1\dots d\tau_J$

$= \int_{0 \leq x_1,\dots,x_J; \sum x_j \leq 1/\tilde\delta - J} dx_1\dots dx_J$

$= \frac{1}{J!}(\frac{1}{\tilde\delta} - J)^J.$

2. Some typos:

2a. "d_1, r_1 respectively" should be "d_1, d_2 respectively".

2b. "Proposition 7" should be "Lemma 7".

2c. Exponent $k+1$ should be k_0+1 (at least four occurrences).

2d. Link is missing under "In this previous post".

2e. "exceeds R. or" should be "exceeds R or"



Terence Tao · 22 June 2013

Thanks for the corrections!

The alternate estimate you propose, namely

$$\kappa_3 = \frac{G_{k_0-1,\tilde\theta}(0,0)}{G_{k_0-1}(0,0)} \sum_{0 \leq J \leq 1/\tilde\jmath} \frac{(k_0-1)^J (\frac{1}{\tilde\delta} - J)^J}{(J!)^2}$$

seems to be a bit worse actually than the current estimate

$$\kappa_3 = \frac{G_{k_0-1,\tilde\theta}(0,0)}{G_{k_0-1}(0,0)} \exp(A + (k_0-1)\int_{\tilde\delta}^{\theta} e^{-At}\frac{dt}{t})$$

for (say) v08ltu's most recent values: the latter is about 10^{1954} times the G-ratio but the former seems to be something like 10^{4010}, peaking at J about 4200. Somehow the problem is that $\tilde\delta$ is so small that the double factorial $(J!)^2$ in the denominator has a hard time counteracting the factors of $1/\tilde\delta$ in the numerator. (The current estimate also blows up as $\tilde\delta \to 0$, but the blowup seems to be milder somehow.)



Gergely Harcos · 22 June 2013

Thanks for the information! I haven't experimented with the numerics lately. Another idea is to optimize A for each J individually, i.e. choose A such that $A(1-J\tilde\delta) = J$.



Gergely Harcos · 22 June 2013

Here I meant to use your upper bound for the t-integral, then optimize in A.



Eytan Paldi · 23 June 2013

From my next message (on a sub multiplicative property of $f/f(0)$), it follows that one can take $A = j_{k_0-2}^2/(2k_0-2)$ only in the integral (i.e. without the constant factor $\exp A$)

Gergely Harcos · 23 June 2013

Actually keeping the t-integral (i.e. not bounding it) gives much better results. I also experimented with replacing $G_{k_0-1,\tilde{\theta}}(0,0)$ by $G_{k_0-1,\tilde{\theta}}(J\tilde{\delta},J\tilde{\delta})$ in the J-term. We can gain a factor of $\approx 10^{56}$ this way, but for $k_0=1466$ we would need a gain of $\approx 10^{208}$.

Eytan Paldi · 23 June 2013

[Edited, Jun 25, to incorporate corrections by the author – T.]

A sub-multiplicative property of f and new bounds for the κ's:

1. Let $g\in C^1[0,1]$, be non-vanishing on $[0,1)$ (with the possible vanishing at 1.) Consider the following inequality for $t,\tau,t+\tau\in[0,1]$:

(1) $|g(t+\tau)|\le K(\tau)|g(t)|$

Where for each fixed $\tau\in[0,1)$ $K(\tau):=\sup_{0\le t<1-\tau}|g(t+\tau)/g(t)|$ is (obviously) the best (uniform) multiplier $\forall t\in[0,1-\tau]$.

Clearly, a sufficient condition for $K(\tau)=|g(\tau)/g(0)|$ to hold for each fixed $\tau\in[0,1)$, is that the function $|g(t+\tau)/g(t)|$ should be non-increasing (as a function of t) on $[0,1-\tau)$. That is, its logarithm (as a function of t) should be non-increasing on $[0,1-\tau)$. Or equivalently, the logarithmic derivative of $|\ g(t+\tau)/g(t)$ should be non-positive on $[0,1-\tau)$.

It follows that the above sufficient condition is implied by

(2) $\log|g|$ is concave on $[0,1)$.

To prove the implication, observe that for a fixed $\tau\in[0,1)$, the derivative of $\log|g(t+\tau)/g(t)|$ is $h(t+\tau)-h(t)$ where $h:=|g|'/|g|$ is the logarithmic derivative of g. Now (2) implies that h (as a non-increasing function on $[0,1]$)) satisfies $h(t+\tau)-h(t)\le 0$ whenever $t,\tau,t+\tau\in[0,1)$, which proves the sufficiency of (2) for (1) with $K(\tau)=|g(\tau)/g(0)|$.

This motivates the following

[For some reason, LaTeX is not working on the line below, I was unable to fix it – T.]

Definition: A function g \in C^1[0,1] will be called *nice* (on [0,1]) if it is non-vanishing on [0, 1) and satisfy

(3) $g(0)=1$ and $(\log g)$ is concave on $[0,1)$

Consequently, any nice function g on [0, 1] satisfies

(4) $g(t+\tau)\le g(t)g(\tau)$, whenever $t,\tau,t+\tau\in[0,1]$

(4') $\log g(t)\le g'(0)t$ on $[0,1)$.

Where (4') follows from the concavity of $\log g$ on $[0,1)$.

It is easy to verify the following properties

(i) A finite product of nice functions is nice.
(ii) If g_n is a sequence of nice and analytic functions on
$[0,1]$, converging locally uniformly on $[0,1]$ to a function g, than g is (obviously) analytic and nice on $[0,1]$.

Examples: If $a, \alpha \in \mathbb{R}$ and $a \neq 0$, then the functions

$E_0(t/a), E_1(t/a), \exp(\alpha t)$ are nice, where

$E_0(t) := 1-t, E_1(t) := (1-t)\exp(t)$

Theorem: Let g be an entire function of order < 2, having its zeros in $\mathbb{R} \setminus [0,1)$ and satisfying $g(0) = 1$, than g is a nice function on $[0,1]$ and, consequently, it satisfies properties (4)-(4').

Proof: denote by h, λ the genus and order of g respectively. Since $h \leq \lambda$ we have two possibilities
(i) $h = 0$: implying that the canonical product representation of g
is given by $g(z) = \prod_n E_0(z/a_n)$ where a_n denote
the zeros of g, since (by the first example), each canonical factor
(i.e. $E_0(t/a_n)$) is nice and the canonical product converges locally uniformly to $g(t)$ on $[0,1]$, it follows (from the above properties) that g is nice on $[0,1]$.
(ii) $h = 1$: implying that the canonical product representation of g is given similarly by $g(z) = \exp(\alpha z)\prod_n E_1(z/an)$ with the same proof.

2. Applications:

Let $f(t) := t^{1-k_0/2} J_{k_0-2}(j_{k_0-2}\sqrt{t})$ for
$t \in [0,1]$, then the product representation (9.5.10) in A&S shows that f is an entire function of genus $h = 0$ having its zeros in $[1, +\infty)$.
Hence by the above theorem $f(t)/f(0)$ is nice on $[0,1]$. hence properties (4) and (4') imply

(5) $f(\sum_j t_j)/f(0) \leq \prod_j [f(t_j)/f(0)]$ whenever the t_j's are non-negative and $\sum_j t_j \leq 1$.
Remark: the last limitation on the sum is unnecessary if f is modified to $f = 0$ on $[1, \infty)$.

(5') $f(t)/f(0) \leq \exp[\frac{f'0}{f(0)}t]$ for each $t \in [0,1]$
(with the above modification of $f = 0$ for $t \geq 1$, (5') holds obviously for each $t \geq 1$.

Remark: it is easy to verify that $f'(0)/f(0) = -J^2_{k_0-2}/[4(k_0-1)]$
which for large k_0 is asymptotic to $-k_0/4$.

Using (5), (5') it is easy to verify that for each $t \geq 0$

$G_{k_0-1}(0,t)/G_{k_0-1}(0,0) \leq f(t)/f(0) \leq \exp(-\alpha t)$

Where $\alpha := -f'(0)/f(0) = j^2_{k_0-2}/[4(k_0-1)]$

Since α is asymptotic (for large k_0) to $k_0/4$, it seems that the last exponential bound is inferior to the current bound

$(1-t)_+^{(k_0-1)/2}$ which determines the upper bound on
κ_1, similar situation exists for the new bound

$G_{k_0-1}(t,t)/G_{k_0-1}(0,0) \le [f(t)/f(0)]^2 \le \exp(-2\alpha t)$
relative to the current bound $(1-t)_+^{k_0-1}$ which determines the current upper bound on κ_2.

the situation seems better for the new bound on κ_3:

Using properties (5)-(5') of f, it follows that

$G_{k_0-1,\tilde\theta}(\sum t_j, \sum t_j)/G_{k_0-1,\tilde\theta}(0,0)$
$\le \prod_j [f(\sum t_j)/f(0)]^2 \le \prod_j [f(t_j)/f(0)]^2$
$\le \prod_j \exp(-2\alpha t_j)$

It is easy to verify that the last exponential bound implies the current bound with with the parameter $A = 2\alpha = j_{k_0-2}^2/[2(k_0-1)]$ in the integrand, but without the constant multiplier $\exp(A)$.

Gergely Harcos · 23 June 2013

These are very good ideas. Some thoughts:

1. Your (4) and (4′) can be proved directly for

$f_k(t) := t^{-k/2} J_k(j\sqrt{t})$ with $j > 0$ fixed,

divided by $f_k(0)$ initially to make its value 1 at $t = 0$. All we need is that $(\log f_k)'$ is decreasing on $(0,\infty)$, i.e. that $(f_k')^2 - f_k f_k'' \ge 0$ on $(0,\infty)$. We have that $f_k' = (-j/2) f_{k+1}$ and $f_k'' = (j^2/4) f_{k+2}$, so we need

$J_{k+1}^2(x) - J_k(x) J_{k+2}(x) \ge 0$ for $x > 0$.

This in turn follows from Auluck's formula that you quoted earlier:

$J_{k+1}^2(x) - J_k(x) J_{k+2}(x) = \frac{2}{x^2} \int_0^x J_{k+1}^2(t)\, t dt.$

2. So we have now a direct proof that

$G_{k_0-1,\tilde\theta}(\sum t_j, \sum t_j)/G_{k_0-1,\tilde\theta}(0,0) \le \prod_j \exp(-2\alpha t_j),$

where $2\alpha = j_{k_0-2}^2/(2(k_0-1))$. This means that Terry's κ_3 can be improved to

$\kappa_3 = \frac{G_{k_0-1,\tilde\theta}(0,0)}{G_{k_0-1}(0,0)} \exp(A + (k_0-1)\int_{\tilde\delta}^{\tilde\theta} e^{-(A+2\alpha)t} \frac{dt}{t})$

for any $A > 0$.

Gergely Harcos · 23 June 2013

Well, perhaps I should have restricted to $k = k_0 - 2$, $j = j_{k_0-2}$, and the interval $(0,1)$ instead of $(0,\infty)$.

Eytan Paldi · 23 June 2013

This is a nice observation! I'm quite surprised to see that the log-concavity holds also for $[1,\infty)$, though at least in a sufficiently small neighborhood of a zero, the dominating logarithmic singularity is concave, the problem is that the logarithmic derivative is positive sufficiently close to the right of each zero, this is exactly the

reason why the interval of interest should not have zeros of f.
Another well known example is $1/\Gamma(t)$ known to be entire, log-concave and monotonically decreasing to the right of its maximizer (which is in $(0,1)$) – in this case the zeros are to the left of the "interesting interval"!



Gergely Harcos · 23 June 2013

The new κ_3 that arises from Eytan's ideas seems to allow $k_0 = 1466$. Using $\varpi = 1/148 - 1/318000$, $\delta' = 1/200$, $A = 1360$, I am getting $\epsilon \approx 8.41 \times 10^{-8}$, $2\kappa_1 \approx 7.15 \times 10^{-8}$, $2\kappa_2 \approx 2.99 \times 10^{-11}$, $2\kappa_3 \approx 3.69 \times 10^{-56}$. These values might not be optimal, and they should be double checked.



Terence Tao · 23 June 2013

Maple seems to confirm it. Nice work by both of you!

.

```
varpi := 1/148 - 1/318000;
k0 := 1466;
deltap := 1/200;
A := 1360;

delta := (1 - 148 * varpi) / 33;
theta := deltap / (1/4 + varpi);

# Gergely's improved value for thetat
thetat := ((deltap - delta)/2 + varpi) / (1/4 + varpi);
deltat := delta / (1/4 + varpi);

j := BesselJZeros(k0-2,1);

eps := 1 - j^2 / (k0 * (k0-1) * (1+4*varpi));

kappa1 := int( (1-t)^((k0-1)/2)/t, t = theta..1, numeric);
kappa2 := (k0-1) * int( (1-t)^(k0-1)/t, t=theta..1, numeric);

# using Gergely and Eytan's improved kappa_3

alpha := j^2 / (4 * (k0-1));
e := exp( A + (k0-1) * int( exp(-(A+2*alpha)*t)/t, t=deltat..theta, numeric ) );

# using Gergely's exact expression for denominator
gd := (j^2/2) * BesselJ(k0-3,j)^2;

# using Eytan's exact expression for numerator
tn := sqrt(thetat)*j;
gn := (tn^2/2) * (BesselJ(k0-2,tn)^2 - BesselJ(k0-3,tn)*BesselJ(k0-1,tn));

kappa3 := (gn/gd) * e;

eps2 := 2*(kappa1+kappa2+kappa3);

# we win if eps2 < eps
```

v08ltu · 23 June 2013

In your Maple worksheet, you integrate, deltat to theta, but the formula listed by Harcos is $\int_{\tilde{\delta}}^{\theta}$.

[Well spotted! But I think Gergely's formula has a typo, the integral here (which is counting primes up to $R^\theta = x^{\delta'}$) should go up to θ rather than θ'. In any case the exponential decay of the integrand renders the upper limit of the integral more or less irrelevant. -T.]

Gergely Harcos · 24 June 2013

Indeed I had a typo, thanks for pointing it out! It was inherited by copy+paste from Terry's earlier comment (https://terrytao.wordpress.com/2013/06/18/a-truncated-elementary-selberg-sieve-of-pintz/#comment-235766), but my SAGE worksheet contained the right endpoint θ. Also, as Terry explained, the figures would still be fine with integration up to ∞.

Gergely Harcos · 24 June 2013

I mean that https://terrytao.wordpress.com/2013/06/18/a-truncated-elementary-selberg-sieve-of-pintz/#comment-235986 was a response to https://terrytao.wordpress.com/2013/06/18/a-truncated-elementary-selberg-sieve-of-pintz/#comment-235937

Terence Tao · 24 June 2013

Oops! Sorry about that; too late, now, but I edited the comment to fix it.

Eytan Paldi · 25 June 2013

By (9.5.4) in A&S, it follows that $J_{n-1}(j_n)/J_{n+1}(j_n) = -1$, so this ratio may be used for $n = k_0 - 2$ to check the relative numerical error.

The distribution of primes in densely divisible moduli | What's new · 23 June 2013

[...] The difference between this conjecture and the stronger conjecture is that the modulus is constrained to be -densely divisible rather than -smooth (note that is no longer constrained to lie in). This relaxation of the smoothness condition improves the Goldston-Pintz-Yildirim type sieving needed to deduce from ; see this previous post. [...]

Terence Tao · 25 June 2013

I wrote a wiki page at

http://michaelnielsen.org/polymath1/index.php?title=Dickson-Hardy-Littlewood_theorems

recording most of the techniques we have for converting results of the form $MPZ[\varpi, \delta]$ or $MPZ'[\varpi, \delta]$ to $DHL[k_0, 2]$. Nothing new on that page, but it at least collects all the different implications that we have found in a single location (though they are all basically obsolete now that we have Pintz's sieve).

 **Bounded gaps between primes (Polymath8) – a progress report | What's new** · 30 June 2013

[...] "dense divisibility", replacing the conjecture by a variant that we have called); see this post for details. Finally, by optimising the remaining steps in Zhang's argument (see below), we [...]

 **Eytan Paldi** · 5 July 2013

The current definition of $\tilde{\theta}$ ensures (under theorem 5 hypotheses) that $\tilde{\theta} < 1$, but without the factor $1/2$ in its definition, we need to add this constraint to theorem 5 hypotheses.

 **Gergely Harcos** · 5 July 2013

I don't think we need this constraint. If we replace $\tilde{\theta}$ by $\min(\tilde{\theta}, 1)$, then the function $G_{k_0-1,\tilde{\theta}}(t_1, t_2)$ does not change, hence the condition of Theorem 5 remains the same, while its proof works verbatim, i.e. the original stated conclusion holds.

 **An improved Type I estimate | What's new** · 27 July 2013

[...] We use the Pintz sieve from this post, repeating the proof of Theorem 5 from that post (and using the explicit formulae for and from [...]